

Om Bhatt

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Education

Texas A&M University

College Station, TX | May 2028

Bachelor of Science in Mechatronics Engineering, Minor in Embedded Systems

Relevant Coursework: Embedded Systems (C), Microcontroller Architecture, Applied Dynamic Systems, Calc I-III, Diff Eq

Skills

Robotics & Autonomy: ROS 2, PX4, MAVROS, SLAM, OpenCV, PWM/PID Control, Teleoperation, Gazebo, RViz, Point Clouds

Hardware and Prototyping: ESP32, Arduino, Raspberry Pi, STM32, FPGA, LiDAR, Cameras, PCB Design, Soldering, Analog Electronics, Oscilloscope & DMM Testing, Data Acquisition Systems

Programming Languages: Python, C/C++, Embedded C, Linux, Shell (Bash/Zsh), LaTeX, HTML, CSS

Simulation and CAD: SolidWorks, FEA, Autodesk Inventor, Onshape, Bambu Studio, Eagle, Multisim, MATLAB/Simulink

AI & Software Tools: Git/GitHub, Visual Studio Code, Workspace, Microsoft Office, LLM Integration (ChatGPT, Claude, Gemini)

Relevant Experience

Undergraduate Research Assistant | *Autonomous Systems Lab, Texas A&M University*

College Station, TX | Aug 2026 – Present

- Researching and implementing algorithms for autonomous ground robot navigation under Dr. Sivakumar Rathinam at the Autonomous Systems Laboratory, J. Mike Walker '66 Department of Mechanical Engineering.

R&D Robotics Engineer | *T.U.R.T.L.E Robotics, Texas A&M University*

College Station, TX | Sep 2025 – Sep 2026

- Researched and programmed autonomous flight algorithms (offboard_control_square_2) using PX4/MAVROS over ROS 2 Humble, validating autonomous flight via 2-drone simulation and 2-3 outdoor test flights for a 10-member team's UAV swarm platform assisting first responders.
- Co-authored and presented the "Disaster Response Observation Network (DRON)" poster at TURTLE Robotics' public Project Showcase (Nov 2025, Apr 2026), one of 21+ advanced projects presented to students, faculty, and the public.

Hardware Team Member | *Texas Aimbots, Texas A&M University*

College Station, TX | Sep 2025 – Sep 2026

- Designed and 3D-printed 4+ ABS structural components in Onshape CAD for Sentry, a fully autonomous ground robot, including a gimbal housing redesigned for a more compact, rigid footprint; installed on the robot.
- Integrated the NVIDIA Jetson Orin, LiDAR, and camera systems to enable real-time perception and autonomous decision-making ahead of the 2026 ARC Championship, an international collegiate robotics competition.

AEOP Summer Robotics Research Intern | *Center for Autonomy, UT Austin*

Austin, TX | Jun 2025 – Aug 2025

- Researched autonomous off-road navigation and drift/jump control dynamics under Dr. Ufuk Topcu and Dr. Christian Ellis as a \$6,000-stipend AEOP REU Scholar to architect a low-cost, 1/16 scale autonomous RC car as a modular research platform for drift and jump navigation studies.
- Executed the complete electromechanical assembly by designing and soldering a custom circuit board to power and integrate all components, including a Raspberry Pi 5/Jetson Nano and motor controllers, verified via multimeter testing.
- Programmed a low-level C++/ROS 2 controller for Ackermann steering and throttle control, validating full teleoperation across 5+ test runs, including a successful jump test.
- Ran a visual SLAM algorithm using camera-based perception in the F1Tenth simulator to validate localization and control, laying the groundwork for planned IMU fusion and post-jump re-localization toward fully autonomous operation.

Mechatronics Research Intern | *Mechatronics Vehicle Systems Lab, University of Waterloo*

Waterloo, ON | Jun 2021 – Sep 2021

- Engineered an external HMI (eHMI): an LED matrix display for WATonoBus, Professor Amir Khajepour's autonomous shuttle at MVS Lab, as 1 of 4 team members (2 students, 1 PhD student) on the eHMI subsystem.
- Developed a ROS node parsing intention-based commands over a rosserial link, validating all icon states and display functions in a live outdoor test on the shuttle's campus loop route.

Projects

T.U.R.T.L.E Robotics Hatchling Robot | *ESP32, Arduino, C/C++, SolidWorks, Kinematics, Actuation & Motor Control* Jan 2025 – Sep 2025

- Won 2nd Place out of 30+ competing teams at the 2025 TURTLE Robotics Hatchling Competition.
- Designed and CADded key mechanical components in SolidWorks, including a forklift-style lift and single-block gripper mechanism, as part of a 4-person team, engineered for rapid sequential block acquisition.
- Programmed an ESP32 and Arduino to control precise vehicle navigation and operate the multi-stage lift, enabling the robot to retrieve and stack 7 blocks in competition.

EcoTrack: Dual-axis Solar Tracker | *ESP32, C/C++, SolidWorks, FEA, Thermo-responsive polymers, Kinematics*

Sep 2022 – Mar 2024

- Researched phototropism and thermo-responsive polymers to engineer a solar tracker with no electrical tracking components, and built an ESP32-controlled benchmark system, outperforming traditional trackers by 87% and fixed-panel systems by 119% in net energy harnessed.
- Achieved 1st place at the Austin Regional Science Fair, advanced to TXSEF State Fair, and won a \$2,000 Jacobs Engineering Scholarship and the RICOH Sustainable Development Award.

Volunteering & Leadership

Autodesk Ambassador Program | *Autodesk, Texas A&M University*

College Station, TX | Jan 2026 – Present

- Selected for a competitive ambassador program to represent Autodesk, promoting CAD tools through campus events and workshops.

TXSEF Science and Engineering Fair Judge | *Texas A&M College of Engineering*

College Station, TX | Mar 2025 – Present

- Evaluate 20+ student research projects annually in the Energy: Sustainable Materials & Designs and Robotics & Intelligent Machines categories at the Texas Science and Engineering Fair (TXSEF).

Design Review Officer | *T.U.R.T.L.E Robotics, Texas A&M University*

College Station, TX | Jan 2026 – Aug 2026

- Evaluated system requirements, CAD/electronics documentation, and BOMs across 20+ projects over 3 Design Review sessions.

Sponsorship Committee Member | *T.U.R.T.L.E Robotics, Texas A&M University*

College Station, TX | Sep 2025 – Jan 2026

- Secured corporate sponsorships for TURTLE through a new Corporate Sponsor Outreach Initiative by developing and executing outreach strategies to engage companies and fund workshops, social events, and advanced technical projects.